

ShoreGuard - Quality Assurance and Testing

QA Overview

This section documents the quality assurance and testing process for the Surf Lifesaving Volunteer Patrol Management App. The purpose of testing is to verify that the Power App meets the acceptance criteria defined in Assignment 1 and works correctly for both volunteer lifeguards and patrol coordinators.

The testing process includes a formal test plan, acceptance criteria mapping, Power Apps Test Studio automated testing, Power Apps Monitor performance evaluation, accessibility evaluation, and documented testing limitations.

B1. Formal Test Plan and Acceptance Criteria Mapping

The purpose of this testing plan is to check whether the Surf Lifesaving Volunteer Patrol Management App works correctly for volunteer lifeguards and patrol coordinators.

Testing focused on realistic patrol tasks such as volunteer registration, shift applications, certification approval, patrol roster management, and urgent surf alerts. Both manual testing and Power Apps Test Studio were used during testing.

Some workflows could not be fully automated because of Power Apps limitations, especially for attachment uploads and modern UI controls. In these cases, screenshots and manual verification were used instead.

The QA testing was based on three main acceptance criteria from Assignment 1. These criteria were used to make sure our testing was linked back to the original project requirements, rather than only testing random screens in the app.

AC1 – Volunteers can create and manage their patrol availability profile.

This criterion checks whether volunteer lifeguards can enter their personal patrol information, update their availability, and make this information available for patrol coordinators. This is important because coordinators need accurate volunteer availability before creating rosters.

AC2 – Coordinators can verify volunteer certifications before assigning shifts.

This criterion checks whether patrol coordinators can review volunteer certification information before assigning volunteers to patrol shifts. Volunteers with valid or approved certification

should be eligible for assignment, while volunteers with missing or expired certification should not be assigned until their status is updated.

AC3 – Coordinators can manage rosters and urgent alerts.

This criterion checks whether patrol coordinators can create or update patrol rosters and send urgent surf alerts to volunteers when needed. This is important because patrol coordination depends on both planned roster management and quick communication during changing beach or safety conditions.

Test ID	Acceptance Criteria	Test Procedure	Expected Result	Actual Result
TC01	AC1 – Volunteers can create and manage their patrol availability profile	Enter volunteer information and submit registration form	Volunteer profile is saved successfully	Pass – profile information saved correctly during testing
TC02	AC1 – Volunteers can create and manage their patrol availability profile	Open patrol roster and apply for an available shift	Shift request appears in coordinator dashboard	Pass – shift application updated correctly
TC03	AC2 – Coordinators can verify volunteer certifications before assigning shifts	Upload certification document and approve volunteer status	Certification status changes to Approved	Pass – updated certification visible in volunteer profile
TC04	AC3 – Coordinators can manage rosters and urgent alerts	Edit roster assignments and save changes	Updated roster is displayed correctly	Pass – roster changes synchronised successfully

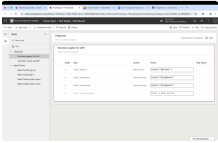
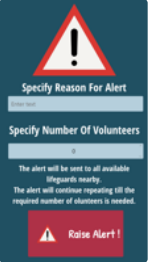
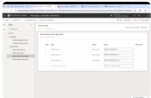
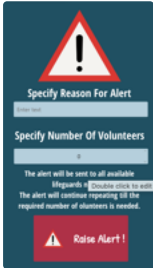
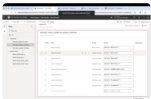
TC05	AC3 – Coordinators can manage rosters and urgent alerts	Send urgent alert notification through app	Alert message becomes visible to volunteers	Partial Pass – alert displayed correctly but some refresh delay was observed
TC06	AC1 / AC2 – Form validation and certification-related data entry are checked	Submit registration form with missing information	Validation message prevents submission	Pass – required fields validated correctly

B2. Power Apps Test Studio Automated Test Suite

Power Apps Test Studio was used where possible to create and run automated tests for the main app workflows. The automated test suite focused on repeatable user actions such as creating a volunteer profile, applying for a shift, assigning a volunteer to a roster, sending an urgent alert, and responding to an urgent alert.

Some functions were not fully supported by Test Studio, especially file upload and some modern Power Apps components. These limitations were documented and supported with manual testing evidence where required.

Test Studio Test Case	Workflow Tested	Automated / Manual	Result	Evidence
TS01_CreateVolunteerProfile	Volunteer creates profile	Automated	Pass – Test Studio recording was created and the profile workflow was checked	

TS02_ApplyForShift	Volunteer applies for shift	Automated	Pass – shift application workflow was recorded and checked	
TS03_AssignVolunteerToRoster	Coordinator assigns volunteer	Automated / Manual	Manual verification – roster assignment was checked manually because this workflow depends on coordinator-side data changes	
TS04_SendUrgentAlert	Coordinator sends urgent alert	Automated	Pass – urgent alert creation workflow was checked	
TS05_RespondToUrgentAlert	Volunteer responds to alert	Automated	Manual verification – volunteer response was checked through the app interface	
TS06_CertificationUpload	Volunteer uploads certification	Manual due to Test Studio limitation	Manual verification – certification upload was tested manually due to Test	

			Studio limitations with file upload controls	
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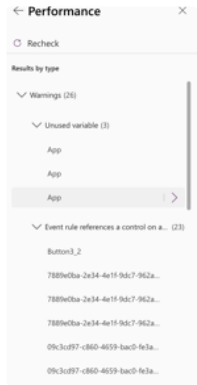
B3. Power Apps Monitor Performance Evaluation

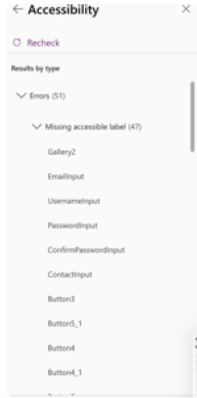
Power Apps Monitor was used while testing the main patrol management screens and volunteer workflows.

During testing, the volunteer dashboard loaded normally with smaller datasets. However, loading became slower when multiple SharePoint lists were requested at the same time.

The largest delay was observed when opening the coordinator roster management screen. In most tests, the roster information took around 2–3 seconds to fully load because several SharePoint network calls were triggered together.

No major runtime crashes were observed during testing. However, some screens occasionally required a manual refresh after SharePoint data updates.

Monitor Area	What Was Checked	Observation	Evidence
App loading	Whether the app opens without errors	Pass – the app opened without a major runtime crash during testing	
Data calls	Whether the app loads data from backend sources correctly	Multiple SharePoint data calls were triggered when opening roster-related screens	
Screen transitions	Whether navigation	Mostly smooth, but the coordinator	

	between screens is smooth	roster screen showed a short loading delay	
Runtime errors	Whether red error entries appear in Monitor	No major red error entries were observed during the test session	
Performance issue	At least one concrete performance behaviour	The coordinator roster screen took around 2–3 seconds to fully load when several SharePoint lists were requested together	

During the Monitor test, the main app workflows were executed, including opening the dashboard, viewing available shifts, submitting a shift application, assigning a volunteer, and sending an urgent alert. The Monitor results will be used to identify whether the app has unnecessary data calls, slow screen transitions, or runtime errors.

B4. Accessibility Evaluation

Accessibility testing focused on whether volunteer lifeguards could quickly use the app during patrol activities, including outdoor environments with bright sunlight and limited attention.

Large buttons and simple navigation were prioritised for important actions such as shift applications and urgent alerts. Testing also checked whether text remained readable on mobile devices during outdoor use.

Most navigation features were easy to use during testing. However, some Power Apps responsive layouts were less consistent on smaller phone screens.

Some accessibility limitations remain because of Power Apps platform constraints, especially when using modern UI components.

Accessibility Area	Check	Result	Comment / Evidence
Text readability	Text is large and clear enough to read	Pass	Main text was readable on desktop and mobile preview
Colour contrast	Text and background are easy to distinguish	Pass	Key pages used clear contrast during normal indoor testing
Button size	Buttons are large enough for mobile use	Pass	Main actions such as Apply, Save, and Alert used large touch targets
Input labels	Forms have clear labels	Pass	Volunteer profile and shift forms used visible field labels
Error messages	Error messages explain what went wrong	Partial Pass	Required field warnings appeared, but some system messages were still generic
Navigation clarity	Users can find core functions easily	Pass	Profile, roster, certification, and alert screens were easy to locate
Field use	App can be used quickly by volunteers during patrol	Partial Pass	Core actions were accessible, but smaller phone screens may require

			further layout refinement
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The app should support volunteers who may need to use it quickly in outdoor conditions. Therefore, important actions such as applying for shifts and responding to urgent alerts should use clear labels and large buttons. Any platform limitations in Power Apps accessibility support were documented in the testing limitations section.

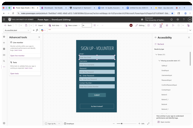
B5. Testing Limitations and Workarounds

Power Apps Test Studio could not reliably automate some workflows involving attachment uploads and modern UI controls.

As a result, several certification upload tests were completed manually and verified using screenshots and backend SharePoint records.

During repeated testing, some automated test results became inconsistent because SharePoint test data changed after earlier test runs. Controlled test records were later introduced to improve repeatability.

Some refresh actions inside the app also behaved differently between desktop and mobile testing, so both environments were checked separately where possible.

Limitation	Impact	Workaround	Evidence
File upload may not be fully supported in Test Studio	Certification upload may not be automated	Manual test with screenshot	
Some modern controls may not record correctly	Some actions may not replay reliably	Manual verification	
Role-based access may require different accounts	Volunteer and coordinator workflows need separate testing	Test with different user roles or test data	

Backend data changes may affect repeatable tests	Test results may change after data updates	Use controlled test records	
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B6. Evidence Summary

Required Evidence	Included?	Location
Formal test plan	Yes	Section B1
Acceptance criteria mapping	Yes	Section B1
Final pass/fail results	Yes	Section B1
Test Studio test suite screenshot	Partially included	Section B2
Test Studio run result screenshot	Partially included	Section B2
Manual testing screenshots	Partially included	Section B1 / B2
Power Apps Monitor screenshot	Pending / to be added before submission	Section B3
Concrete Monitor observation	Yes	Section B3
Accessibility checklist	Yes	Section B4
Testing limitations and workarounds	Yes	Section B5